INCH-POUND

MIL-DTL-82894A(OS)
29 May 2024
SUPERSEDING
MIL-C-82894(OS)
9 February 1994

DETAIL SPECIFICATION

CARTRIDGE, FIRE EXTINGUISHER, CCU-126/A

Reactivated after 29 May 2024 and may be used for new and existing designs and acquisitions.

This specification is approved for use by the Naval Surface Warfare Center Indian Head Division (NSWCIHD) and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the CCU-126/A fire extinguisher cartridge, referred to herein as cartridge.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3, 4, and 5 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3, 4, and 5 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

FEDERAL SPECIFICATIONS

PPP-B-676 - Boxes, Setup

PPP-C-1797 - Cushioning Material, Resilient, Low Density, Unicellular, Polypropylene Foam

COMMERCIAL ITEM DESCRIPTIONS

A-A-1507 - Chipboard

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-DTL-117 - Bags, Heat-Sealable

MIL-DTL-23659 - Initiators, Electric, General Design Specification for

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-129 - Military Marking for Shipment and Storage

Comments, suggestions, or questions on this document should be addressed to Commander, Naval Sea Systems Command, ATTN: SEA 05S, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard, DC 20376-5160 or emailed to CommandStandards@navy.mil, with the subject line "Document Comment". Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

AMSC N/A FSC 1377

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MIL-STD-810	-	Environmental Engineering Considerations and Laboratory Tests
MIL-STD-1168	-	Ammunition Lot Numbering and Ammunition Data Card
MIL-STD-2073-1	-	Standard Practice for Military Packaging

(Copies of these documents are available online at https://quicksearch.dla.mil.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

NAVAL AIR SYSTEMS COMMAND (NAVAIR) DRAWINGS

834AS450	-	Cartridge, Aircraft Fire Extinguisher CCU-l26/A Assembly, Loaded
834AS471	-	Ignition Charge
838AS102	-	Powder, Smokeless

(Copies of these drawings are available from Commander, Indian Head Division, Naval Surface Warfare Center, CAD/PAD Data Management Branch (Code E26), 4393 Benson Road, BLDG 1557, Indian Head, MD 20640-5092. Refer questions, inquires, or problems to nswc_iheodtd_dmo@navy.mil.)

NAVAL SUPPLY SYSTEMS COMMAND INSTRUCTIONS

NAVSUPINST 4030.28	-	Logistics, Packaging of Materiel

(Copies of this document are available online at https://nll.navsup.navy.mil.)

CODE OF FEDERAL REGULATIONS (CFR)

49 CFR 171-178	-	Hazardous Materials Regulations, parts 171-178

(Copies of this document are available online at www.ecfr.gov.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D6880/D6880M	-	Standard Specification for Wood Boxes
ASTM E748	-	Standard Guide for Thermal Neutron Radiography of Materials
ASTM E1742/E1742M	-	Standard Practice for Radiographic Examination

(Copies of these documents are available online at www.astm.org.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.2.

3.2 Flight critical part. The CCU-126/A fire extinguisher cartridge (Drawing 834AS450) is a flight critical item and shall be identified and produced as such.

3.3 Design and construction. The CCU-126/A fire extinguisher cartridge covered by this specification shall be manufactured in accordance with Drawing 834AS450 and all documents listed thereon.

3.4 Materials. All materials used in the manufacture of the cartridges shall conform strictly to the respective drawings. Any departures from the drawings require written approval by the Technical Authority (see 6.4) prior to manufacture.

3.5 Primary components. The ignition charge (Drawing 834AS471) and the smokeless powder (Drawing 838AS102) are primary components. Only one lot of each primary component shall be used in a cartridge lot. One primary component lot may be used in more than one cartridge lot.

3.6 Cartridge requirements.

3.6.1 Visual. Each cartridge shall be free of visible defects and have proper identification markings as defined in Drawing 834AS450 when inspected in accordance with 4.5.1.

3.6.2 Bridge circuit resistance. Each cartridge's bridge circuit resistance shall be 1.00±0.15 ohms when measured in accordance with 4.5.2.

3.6.3 Radiographic inspection. Each cartridge shall be free of defects or foreign material and shall have proper assembly, presence of parts, proper crimping and sealing, and proper electrical connections. Film interpretation of the radiographs shall verify the presence and correct positioning of all parts and explosives (see 4.5.3).

3.6.4 Leakage. The leakage rate of each cartridge shall not exceed 1.0×10^{-5} cc/sec of air when tested in accordance with 4.5.4.

3.6.5 No-fire. The cartridge shall not fire when subjected to a direct current of one ampere at one watt of power for 5 minutes minimum in accordance with 4.5.5.

3.6.6 Static discharge. The static discharge shall not fire or dud the cartridge when tested in accordance with 4.5.6.

3.6.7 Stray voltage. Stray voltage shall not fire or dud the cartridge when tested in accordance with 4.5.7.

3.7 Environmental requirements.

3.7.1 Shock. The cartridge shall not incur damage from the shock test of 4.5.8.

3.7.2 Temperature, shock, humidity, and altitude. The cartridge shall not incur damage or deformation from the temperature, shock, humidity, and altitude test of 4.5.9.

3.7.3 Vibration. The cartridge shall not incur damage or deformation from the vibration test of 4.5.10.

3.7.4 Salt fog. The cartridges shall not incur detrimental deformations including corrosion, peeling, chipping, or blistering of the finish, nor exposure of the base metal, as a result of the salt fog test of 4.5.11.

3.8 Function. The cartridge, when tested in accordance with 4.5.12, shall provide sufficient power to fully rupture the retaining disc of the container, permitting full and rapid discharge of the fire extinguishing agent. There shall be no mechanical failure or distortion that would impair removal of the cartridge.

3.8.1 Ignition time. The cartridge shall fire within 25 milliseconds with application of 5.00, +0.25, -0.00 amps, when tested in accordance with 4.5.12.1.

3.8.2 Misfire. There shall be no misfire.

3.9 Workmanship. The cartridge shall be constructed and finished in a manner to assure compliance with all requirements of this specification. Particular attention shall be directed to dimensions, finishes, crimping, sealing, and assembly operations.

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4. VERFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2).

b. Production inspection (see 4.3).

c. Lot acceptance inspection (see 4.4).

4.2 First article. Unless otherwise specified (see 6.2), a first article sample of 33 cartridges shall be subjected to first article testing. Thirty of these cartridges shall be expended in the tests listed in table I and three cartridges shall be retained for investigative purposes. Any damage inflicted by the environmental treatments, that would adversely affect the performance of the item in service application, shall be cause for rejection of the first article sample. Any further production prior to notification by the contracting agency of first article sample acceptability is at the contractor's risk. Failure of any cartridge to comply with the requirements of table I shall be cause for rejection of the first article represented.

TABLE I. First article test plan.

Test Sequence	Requirement Paragraph	Test Paragraph	Sample Group					
			I	II	III	IV	V	VI 1/
1. Visual	3.6.1	4.5.1	6	6	6	6	6	3
2. Bridge circuit resistance	3.6.2	4.5.2	6	6	6	6	6	3
3. Radiographic	3.6.3	4.5.3	6	6	6	6	6	3
4. Leakage	3.6.4	4.5.4	6	6	6	6	6	3
5. No-fire 2/	3.6.5	4.5.5	6					
6. Static discharge 2/	3.6.6	4.5.6	6					
7. Stray voltage 2/	3.6.7	4.5.7	6					
8. Shock 3/	3.7.1	4.5.8		6	6	6	6	
9. TSHA 3/	3.7.2	4.5.9		6	6	6		
10. Vibration 3/	3.7.3	4.5.10		6	6			
11. Salt fog 3/	3.7.4	4.5.11		6				
12. Function test: -65 °F 70 °F 200 °F	3.8	4.5.12	 2 2 2	 2 2 2	 2 2 2	 2 2 2	 2 2 2	

NOTES:

1/ Investigative sample. Any investigative sample cartridge used shall pass the inspections as shown in the table before being subjected to any other tests.

2/ A bridge circuit resistance test shall be conducted following specified test.

3/ A visual, bridge circuit resistance, leakage, and radiographic inspection shall be conducted following specified test.

4.3 Production inspection. The entire lot of production cartridges manufactured under the contract shall be inspected and screened for defects as defined in table II. Cartridges failing to meet the requirements referenced in table II shall be rejected and removed from the lot.

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TABLE II. Production inspection.

Test	Requirement Paragraph	Test Paragraph
Visual	3.6.1	4.5.1
Bridge circuit resistance	3.6.2	4.5.2
Radiographic	3.6.3	4.5.3
Leakage	3.6.4	4.5.4

4.4 Lot acceptance inspection. Lot acceptance shall consist of the examinations and tests specified in table III. Failure of any sample cartridge to comply with the requirements of table III shall be cause for rejection of the lot represented.

TABLE III. Lot acceptance inspections and tests.

Inspection/Test	Requirement Paragraph	Test Paragraph	Quantity
Visual inspection	3.6.1	4.5.1	Test & retain sample
Bridge circuit resistance	3.6.2	4.5.2	Test & retain sample
Radiographic inspection	3.6.3	4.5.3	Test & retain sample
Leakage test	3.6.4	4.5.4	Test & retain sample
No-fire test	3.6.5	4.5.5	Test sample
Function test: -65 °F 70 °F 200 °F	3.8	4.5.12	 ⅓ Test sample [1/] ⅓ Test sample ⅓ Test sample
NOTE: [1/] Odd cartridges shall be conditioned at -65 °F.			

4.4.1 Sample size. A random sample from each production lot, including the retained sample for investigative purposes, shall be selected in accordance with table IV for lot acceptance inspection.

TABLE IV. Lot acceptance sampling.

Lot Size	Test Sample Size	Retained Sample Size
2 through 50	8	2
51 through 90	13	2
91 through 150	20	2
151 through 280	32	2
281 through 500	50	2
501 through 1200	80	2

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4.5 Inspection and tests.

4.5.1 Visual inspection. Each cartridge produced under the contract shall be visually inspected for proper construction and finishing as specified in 3.7. Particular attention shall be directed to dimensions, finishes, crimping sealing, and assembly operations. The cartridge shall be free of visible defects as noted in 3.6.1.

4.5.2 Bridge circuit resistance. The resistance of each cartridge, from pigtail terminal to case, shall be measured with an accuracy of one percent using equipment which subjects the bridge circuit to a current of 5 milliamperes maximum.

4.5.3 Radiographic inspection. Each assembled cartridge shall be x-rayed and n-rayed in accordance with ASTM E1742/E1742M and ASTM E748, respectively. Any observable imperfections in assembly shall be cause for rejection. The cartridge shall be positioned on its side for the most revealing exposure. All cartridges shall be identified with serial numbers prior to x-ray and n-ray examination. The cartridges for examination shall be arranged on trays or boards in consecutive numerical order by serial number, and each radiographic plate shall carry a permanent identification of the items displayed thereon. The radiographic plate identification shall include the NAVAIR drawing number, the complete lot number, contract number, and the span of serial numbers displayed. Missing serial numbers shall be identified. Radiographic plates of the entire cartridge production lot shall accompany the ballistic sample to the activity conducting the production lot acceptance test (see 4.3). Results shall be examined for conformance to 3.6.3.

4.5.4 Leakage test. Each cartridge in the production lot shall be leak tested in a dry gas leak tester.

4.5.5 No-fire test. A direct current of not less than one ampere supplying a minimum of 1 watt of power shall be applied from the pigtail terminal to the case for a period of 5.0, +0.5, -0.0 minutes.

4.5.6 Static discharge. The cartridge shall be subjected to a static discharge test by being connected to a 500±5 percent picofarad capacitor charged to 25,000±500 volts and a 5,000±5 percent ohm resistor connected in a 500-microhenry total inductance series circuit. The cartridge shall be subjected to the static discharge between the pigtail terminal and the case.

4.5.7 Stray voltage. The cartridge shall be subjected to 2,000 pulses of direct current from the pigtail terminal to the case. Each pulse shall be 300 milliseconds applied at a rate of two per second. Each pulse shall have a minimum amplitude of 100±5 milliamperes. The cartridges shall be conditioned for a minimum of 12 hours at 70±5 °F prior to testing.

4.5.8 Shock test. Cartridges shall be mounted in the device of intended use or equivalent. The testing configuration degree of support shall be the same as that afforded by the device of intended application. The shock pulse shall be applied to the initiator's mounting points in both directions along each of the three mutually perpendicular axes. The shape of each shock pulse shall approximate as nearly as possible a half sine wave. The amplitude of each shock pulse shall exceed 200 g's for 1.5±0.4 milliseconds and it shall exceed 65 g's for 9.0±0.9 milliseconds.

4.5.9 Temperature-shock/humidity/altitude (TSHA). The TSHA test shall be conducted in accordance with the temperature-shock/humidity/altitude test of MIL-DTL-23659.

4.5.10 Vibration test. The cartridge shall be subjected to vibration testing in accordance with MIL-STD-810, Method 514.2-II, Curve H, except that each resonant and cycling period shall be divided equally at -65 °F, 70 °F, and 200 °F.

4.5.11 Salt fog test. Six cartridges shall be subjected to the salt fog test in accordance with Procedure 1 of MIL-STD-810. The cartridges shall be exposed to a 5 percent salt solution for a 48-hour period while in a hanging position.

4.5.12 Function test. Cartridges shall be conditioned for a period of not less than 4 hours nor more than 24 hours at the specified temperatures. The cartridges shall be removed from the conditioning chamber and test fired within 5 minutes. If any cartridge is not fired within 5 minutes after removal, it shall be reconditioned at the specified temperature for an additional 4 hours and then tested. The cartridges shall be initiated using a current of 5.00, +0.25, and -0.00 amperes. The cartridge shall be fired in the Aircraft Fire Extinguisher Container Assembly.

4.5.12.1 Ignition time. The ignition time, which is defined as the time from the first application of current until the first sign of loss of circuit continuity, shall meet the requirements of 3.8.1.

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4.5.12.2 Misfires. A cartridge that does not fire under the conditions specified in 4.4 shall be considered defective, and the lot considered unacceptable, unless the misfire is clearly attributable to faulty test equipment or procedure.

4.5.12.3 Retest. There shall be no retests.

4.5.12.4 Tests failure. If the test failure is attributed to any specific cause excluding the cartridge, the original test results shall be discarded and that test re-conducted.

4.3.13 Inspection of packaging. Packaging, packing, and marking shall be visually examined for conformance to section 5.

5. PACKAGING

5.1 Preservation, packaging, and packing. Preservation, packaging, and packing shall be Level A and shall be in accordance with 49 CFR 171-178 and NAVSUPINST 4030.28. All applicable regulations shall be followed.

5.1.1 Level A packaging. Level A preservation and packaging shall be as specified in a or b as follows:

a. Place each cartridge in one barrier bag conforming to MIL-DTL-117, Type I, Class E, Style 1, size 4 x 6 inches. Evacuate excess air and seal.

b. Package two cartridges in a container conforming to PPP-B-676, Type I, Variety 1, Class C, Style 4, size 5¼ x 4⅜ x 1¼ inches with two 1⅝ x 3¼-inch cells and 7/16-inch air cells on all sides formed by partitions conforming to A-A-1507. Add cushioning per PPP-B-676, Type II or equivalent if required to establish a tight fit. Close container and apply label.

5.1.2 Level A packing. Initiators, packaged as described in 5.1.1, shall be packed in a wooden box conforming to ASTM D6880/D6880M, Class II, Style 2, Grade A, size 18 x 10 x 11¼ inches (interior dimensions). Unless otherwise specified (see 6.2), the number of inner containers packed in the outer container shall be at the option of the supplier. Material conforming to PPP-C-1797 shall be used to fill voids and ensure a tight fit. The box shall be closed, reinforced, and packed tightly according to ASTM D6880/D6880M.

5.2 Marking.

5.2.1 Standard markings. Unless otherwise specified (see 6.2), the marking information of the inner and outer containers shall be as specified below:

a. Inner container marking
   (1) NSN 1377-01-377-9215-SP84
   (2) Cartridge, Aircraft Fire Extinguisher CCU-126/A
   (3) NAVAIR Drawing 30003-834AS450
   (4) Quantity
   (5) Lot number in accordance with MIL-STD-1168
   (6) DOT Explosive Class 1.4

b. Outer container marking
   (1) NSN 1377-01-377-9215-SP84
   (2) Cartridge, Aircraft Fire Extinguisher CCU-126/A
   (3) NAVAIR Drawing 30003-834AS450
   (4) Quantity
   (5) Gross weight and cube
   (6) Lot number in accordance with MIL-STD-1168
   (7) Contract/purchase order number
   (8) WARNING: CARTRIDGES, POWER DEVICES
       UN#0323

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Additionally, the above specified marking information shall be applied to the containers in accordance with the applicable provisions of MIL-STD-129.

5.2.2 Special markings. Additional special markings of outer containers shall be in accordance with 49 CFR 171-178.

5.2.3 Bar coding. Barcode markings shall be in accordance with MIL-STD-129.

5.3 Explanation of terms. For the purpose of this specification, NAVSUPINST 4030.28, when applicable, shall be used to clarify packaging terms.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The cartridge is intended for use in releasing fire extinguishing agent into the area surrounding an aircraft engine in the event of fire. This cartridge is used in the C-130 series fire extinguishing systems.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. When first article inspection is required (see 3.1 and 4.2).

c. Number of inner containers per pack if not optional (see 5.1.2).

d. Marking if other than as specified (see 5.2.1).

e. That the safety requirements of DoD Manual 4145.26, DoD Contractor's Safety Manual for Ammunition and Explosives, are applicable (see 6.6).

6.3 First article. When a first article inspection is required, the contracting officer should provide specific guidance to offerors whether the item(s) should be a preproduction sample, a first article sample, a first production item, a sample selected from the first production items, standard production item from the contractor's current inventory (see 3.1), and the number of items to be tested as specified in 4.2. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.4 Technical Authority. The Technical Authority has the authority, responsibility, and accountability to establish, monitor, and approve technical standards, tools, and processes in conformance to higher authority policy, requirements, architectures, and standards. NAVSEA 05S may be contacted to determine the appropriate Technical Authority for any given requirement related to equipment and systems of U.S. Navy ships and submarines.

6.5 Loaded assembly hazard information.

DoD, DOT, NATO Hazard Classification – Class 1, Division 4
United Nation Serial Number – 0323
Storage Compatibility Grouping – S
DoD Markings – Cartridge, Aircraft Fire Extinguisher CCU-126/A

6.6 Safety precautions. The safety precaution requirements of DoD Manual 4145.26, DoD Contractor's Safety Manual for Ammunition and Explosives, are applicable and should be specified in the contract as required by the Federal Acquisition Regulation (FAR) 23.3.

NOTE: When this specification is used as part of the description of work to be accomplished by a Government activity, the safety requirements of NAVSEA OP 5, Ammunition and Explosives Safety Ashore, are applicable.

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6.7 Subject term (key word) listing.

Aircraft engine compartment
Cartridge activated devices
Electric initiator
Fire extinguishing system
Ignition charge

6.8 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

CONCLUDING MATERIAL

Preparing activity:
Navy – OS
(Project 1377-2024-001)

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